Calibration of Classifier Probability Outputs

1 Calibration as a Measure of Trustworthiness

A classifier can achieve high accuracy while its predicted probabilities still fail to represent its uncertainty reliably. Accuracy measures the proportion of correct predictions, but does not show whether the confidence assigned to those predictions matches their empirical correctness rate. This property is known as *confidence calibration*. A model is calibrated when its estimated probability of being correct agrees with the observed proportion of correct predictions.

Let a classifier produce a probability vector $\mathbf{p}_i = (p_{i1}, \ldots, p_{iK})$ for input x_i. Its predicted class is $\hat{y}_i = \arg\max_k p_{ik}$, and the confidence of that prediction is $c_i = \max_k p_{ik}$. Top-label calibration requires that predictions with confidence c satisfy

$$P(\hat{Y} = Y \mid C = c) = c. \tag{1}$$

This definition has a group-level interpretation: among predictions with confidence close to 0.8, approximately 80 % should be correct. Calibration therefore does not guarantee the correctness of any individual prediction; it describes agreement between estimated confidence and the frequency of correct predictions within a group.

Calibration contributes to trustworthiness by giving users and downstream systems a more meaningful indication of when a model's output may be relied upon. Confidence can, for example, be used to set a threshold below which a prediction is referred for expert review, or to route cases between automated processing and human review. Such uses are relevant in areas including medical diagnosis and autonomous systems, where it can be important to identify confident but incorrect predictions [1, 3]. Calibration is only one aspect of trustworthiness: on its own, it does not establish a model's accuracy, robustness, fairness, or ability to recognize inputs outside the training distribution. An aggregate calibration score may also conceal differences between classes or groups of examples.

1.1 Calibration Evaluation

Calibration can be visualized with a *reliability diagram*. Predictions are grouped into confidence intervals, and within each interval the mean confidence is compared with the proportion of correct predictions. The horizontal axis shows mean confidence, and the vertical axis shows accuracy within the interval. A perfectly calibrated model lies on the diagonal $y = x$. Points below the diagonal indicate overconfidence in that region; points above it indicate underconfidence.

The *Expected Calibration Error* (ECE) is commonly used to summarize the discrepancy. For n examples divided into M bins, let B_m denote the indices of examples in bin m, $\mathrm{acc}(B_m)$ the proportion of correct predictions in that bin, and $\mathrm{conf}(B_m)$ their mean confidence. Then

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{n} \left|\mathrm{acc}(B_m) - \mathrm{conf}(B_m)\right|. \tag{2}$$

The bin weight accounts for how many examples it contains, so a discrepancy in a populated bin contributes more than the same discrepancy in a sparsely populated bin. A lower ECE indicates a smaller average mismatch between confidence and accuracy under the chosen binning scheme.

ECE is a useful and interpretable approximation, but it does not fully characterize calibration. Its value depends on the number and boundaries of the bins, and different settings can produce different results. The standard top-label version also evaluates confidence only for the predicted class, rather than the quality of every probability in the output vector. Subsequent work has proposed alternatives and extensions, including adaptive and class-conditional metrics, which can be more suitable when predictions are unevenly distributed across bins or when there are many classes [2]. ECE can also be complemented by scores such as negative log-likelihood (NLL) or the Brier score. These provide another view of probability quality, although neither is an isolated measure of calibration [1, 5].

1.2 Research Developments and Applications

An influential study of calibration in modern deep neural networks is the work by Guo et al. [1]. The authors showed that higher classification accuracy does not automatically imply better-calibrated outputs. Their experiments on image and document classification tasks found that calibration is affected by architectural and training choices, including network depth and width, batch normalization, and weight decay. They also compared several post-hoc probability calibration methods and found that *temperature scaling* often

performed well on the datasets they studied. It adjusts logits z_{ik} using a single positive parameter T:

$$p_{ik}^{(T)} = \frac{\exp(z_{ik}/T)}{\sum_{j=1}^{K} \exp(z_{ij}/T)}. \tag{3}$$

The parameter T is estimated on a separate calibration set, typically by minimizing NLL. Since $T > 0$, the ordering of the logits is unchanged, so the predicted class remains the same while the sharpness of the probability distribution changes. This illustrates how calibration can be adjusted after classifier training without changing its learned weights.

Later research has shown that calibration should not be assessed using only one metric or only data similar to the training set. Nixon et al. highlighted limitations of ECE and showed that the choice of metric can affect the ranking of calibration methods [2]. Ovadia et al. studied predictive uncertainty under dataset shift and found that post-hoc calibration learned on data from the original distribution may not preserve confidence quality after the inputs change. In their experiments, some methods that account for multiple models were more robust [3]. A later analysis of modern image classifiers found that the relationship between architecture, accuracy, and calibration is not universal: some newer architectures were well calibrated, and trends observed in earlier models were less pronounced [4].

Calibration is therefore used as part of a broader assessment of predictive uncertainty. It is relevant when setting policies for abstaining from a prediction or referring it to a human, but evidence of good calibration on one test set does not automatically extend to other classes, subgroups, or changed data conditions. Results should therefore be reported together with the calibration metric, its settings, and the dataset on which it was computed.

2 Implementation of the Calibration Module

In the implementation, calibration is evaluated as a property of a trained classifier's probability outputs. The experiment in `experiments/calibration.py` loads the CIFAR-10 test split and the ResNet-18 model weights. The `evaluate` function in `evaluation/evaluator.py` switches the model to evaluation mode and obtains its logits without computing gradients. It then applies softmax to obtain class probabilities, selects the predicted class as the index of the largest probability, and records top-label confidence as that maximum probability. It also returns the ground-truth labels so that confidence can be compared with empirical accuracy.

The `calculate_ece` function in `evaluation/calibration.py` divides confidence values into ten equally spaced bins from 0 to 1. For each non-empty bin, it computes the number of examples, mean confidence, and accuracy; the final bin also includes the value 1. The resulting ECE is the weighted average of the absolute differences between accuracy and confidence, as defined in Equation (2). The experiment also creates a reliability diagram that compares bin accuracy with mean confidence and the diagonal representing perfect calibration.

The result is printed to the console and saved to `results/calibration.json`. The JSON file records the model and dataset names, ECE, the number of bins, and, for each non-empty bin, its boundaries, example count, accuracy, and mean confidence. The reliability diagram is saved to `results/reliability_diagram.png`. The scalar metric supports comparisons between runs, while the diagram helps identify regions with the largest mismatch between confidence and accuracy.

The implementation evaluates the calibration of the unadjusted softmax outputs; it does not modify the model parameters or recalibrate its confidence scores. Specifically, it computes top-label ECE using ten fixed bins, so results should be interpreted together with this setting and the reliability diagram. Temperature scaling, described in Equation (3), could be added as a separate step. If parameter T is estimated, a separate calibration set should be used, leaving the test set reserved for final evaluation.

References

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